

RHINELANDER, WI, USA

WMO: 727415

Lat: 45.631N

Lon: 89.482W

Elev: 1624

StdP: 13.85

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 04803

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.3	-9.8	-21.7	1.8	-13.7	-17.2	2.3	-9.4	20.6	17.1	18.9	15.7	5.3	260	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.7	86.1	69.9	82.9	68.2	80.8	66.8	73.1	81.5	71.1	79.2	69.3	77.0	9.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.4	118.8	77.0	68.3	110.5	75.3	66.3	103.0	74.0	37.6	81.5	36.0	79.2	34.4	77.5	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.7	17.4	15.7		-23.1	90.9	3.2	3.4	-25.4	93.4	-27.3	95.4	-29.1	97.3	-31.4	99.8	
				-23.6	76.3	3.1	2.4	-25.8	78.0	-27.6	79.4	-29.3	80.8	-31.6	82.5	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.9	13.6	16.5	27.8	41.1	53.9	62.7	67.3	64.8	57.7	44.6	31.9	18.8	
	DBStd	21.28	12.43	12.10	12.50	9.98	9.09	7.24	6.08	5.93	8.64	8.95	10.19	11.63	
	HDD50	4890	1129	939	697	299	62	4	0	0	26	220	547	967	
	HDD65	8742	1594	1359	1154	719	360	125	46	76	249	635	993	1432	
	CDD50	1917	0	0	8	31	184	385	537	460	256	53	3	0	
	CDD65	295	0	0	0	1	16	57	118	71	30	2	0	0	
	CDH74	2885	0	0	1	15	229	558	1190	622	254	16	0	0	
	CDH80	687	0	0	0	1	55	126	328	130	47	0	0	0	
Wind	WSAvg	7.0	7.6	7.5	7.3	7.8	7.5	6.5	5.9	5.5	6.3	7.2	7.5	7.4	
Precipitation	PrecAvg	31.7	1.1	0.9	1.7	2.6	3.7	4.1	4.0	3.5	3.9	3.0	1.9	1.3	
	PrecMax	42.4	2.6	2.0	3.5	6.0	6.9	7.0	7.8	6.3	8.7	5.6	5.0	3.0	
	PrecMin	24.1	0.2	0.0	0.3	0.7	1.1	1.1	0.9	1.6	0.8	0.9	0.4	0.3	
	PrecStd	4.7	0.6	0.5	0.9	1.2	1.4	1.5	1.6	1.3	1.9	1.4	0.9	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.8	49.7	68.3	76.3	86.1	88.1	90.4	88.0	84.5	76.0	62.7	48.1	
		MCWB	36.5	42.3	56.9	58.3	66.6	71.7	73.3	71.5	70.0	62.2	51.4	44.4	
	2%	DB	36.9	42.5	58.4	69.9	80.4	83.3	87.0	83.5	80.9	70.2	55.7	40.8	
		MCWB	33.7	37.6	49.1	53.7	62.6	67.3	71.0	69.3	67.1	60.3	49.2	37.9	
	5%	DB	33.9	37.0	51.6	63.8	75.3	80.6	83.5	80.6	76.7	64.7	51.5	36.5	
		MCWB	32.1	33.6	44.0	49.1	59.7	66.5	69.2	67.5	64.6	56.2	45.6	33.9	
	10%	DB	30.5	33.7	45.5	59.0	71.0	76.9	81.0	77.2	72.6	60.0	46.4	33.9	
		MCWB	28.9	30.8	39.7	46.9	57.7	63.8	67.4	65.1	63.1	53.1	41.6	32.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.9	43.7	58.5	60.9	70.2	74.1	77.1	74.7	72.2	67.8	54.9	45.6
			MCDB	40.2	48.2	65.8	71.8	81.5	83.0	86.2	82.5	79.7	71.8	58.8	47.1
2%		WB	34.4	37.7	50.9	56.1	66.1	71.0	73.5	72.1	69.4	61.3	50.5	38.0	
		MCDB	36.5	41.6	55.7	66.0	74.9	79.9	82.2	79.3	76.9	67.6	55.6	40.2	
5%		WB	32.0	33.3	44.2	51.4	62.8	68.7	71.4	70.0	67.5	57.5	45.8	34.2	
		MCDB	34.0	36.3	50.5	61.4	71.5	76.7	80.1	76.9	74.1	63.6	49.8	36.0	
10%		WB	29.2	29.9	39.3	47.4	59.6	66.4	69.3	68.0	64.9	53.5	42.2	32.1	
		MCDB	30.8	32.3	44.9	58.5	68.0	73.4	77.6	74.8	70.8	58.7	46.5	33.7	
Mean Daily Temperature Range	5% DB	MDBR	16.0	18.5	19.4	21.0	23.0	21.8	22.7	21.8	21.4	18.1	14.5	14.1	
		MCDBR	16.2	18.9	23.7	30.4	28.4	25.0	25.6	24.3	24.8	24.3	22.4	15.9	
		MCWBR	13.8	14.6	15.1	17.3	13.5	12.4	12.1	12.0	12.8	15.0	15.7	13.1	
	5% WB	MCDBR	15.3	17.7	20.8	25.9	23.1	21.2	21.9	20.5	20.2	21.0	19.9	13.6	
		MCWBR	13.4	13.9	14.1	16.9	12.3	12.1	11.9	11.4	11.9	14.9	15.4	11.7	
Clear-Sky Solar Irradiance	taub	0.240	0.268	0.309	0.335	0.364	0.365	0.362	0.362	0.342	0.306	0.279	0.248		
	taud	2.383	2.297	2.346	2.417	2.377	2.397	2.425	2.425	2.481	2.574	2.541	2.439		
	Ebn at Noon	283	292	293	293	286	286	285	281	279	277	266	268		
	Edh at Noon	23	30	34	35	37	37	35	34	30	23	20	19		
All-Sky Solar Radiation	RadAvg	480	827	1212	1453	1707	1846	1907	1614	1257	777	460	338		
	RadStd	47	88	119	142	144	149	101	105	83	80	47	37		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page